

SURGUT, Russian FederationWMO: **238490**Lat: **61.3424N**Lon: **73.4220E**Elev: **184**StdP: **14.60**Time Zone: **5.00 (E05)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-40.3	-35.2	-46.8	0.4	-40.0	-41.6	0.5	-35.0	22.0	4.0	19.9	1.8	4.6	70	0.745

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.4	83.8	65.5	80.4	64.2	76.8	63.3	68.4	77.7	66.7	75.7	64.9	73.9	10.1	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.5	94.7	71.8	63.0	86.8	70.8	61.0	80.7	69.4	32.9	77.3	31.4	75.3	30.0	73.9	75.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.1	19.4	17.2	DB	-44.2	88.4	7.9	2.5	-50.0	90.2	-54.6	91.7	-59.0	93.1	-64.8	94.9	(4)
(5)			WB	-43.5	71.0	6.6	2.1	-48.3	72.5	-52.2	73.7	-55.9	74.9	-60.7	76.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	29.4	-5.5	-0.3	15.7	28.7	43.1	58.5	63.9	57.7	46.1	32.0	10.1	0.1		
	DBStd	26.95	17.02	15.61	12.83	12.46	11.19	9.89	7.48	7.20	8.02	10.15	16.01	16.90		
	HDD50	8632	1720	1408	1062	642	277	34	1	12	171	559	1198	1548		
	HDD65	13166	2185	1828	1527	1090	682	241	115	247	568	1022	1648	2013		
	CDD50	1096	0	0	0	2	64	290	432	251	54	3	0	0		
	CDD65	154	0	0	0	0	5	47	80	21	1	0	0	0		
	CDH74	1325	0	0	0	0	63	438	676	144	4	0	0	0		
	CDH80	302	0	0	0	0	11	107	164	20	0	0	0	0		
(14) Wind	WSAvg	8.7	7.8	7.9	9.4	9.9	9.8	8.9	8.1	8.0	8.5	9.1	8.7	8.0		
(15) Precipitation	PrecAvg	32.4	2.1	1.7	2.3	2.1	2.5	3.1	2.8	3.5	2.9	3.4	3.3	2.6		
	PrecMax	40.6	4.0	3.3	5.9	3.3	5.9	5.4	6.7	5.9	5.7	5.9	5.4	5.3		
	PrecMin	25.7	0.6	0.4	0.1	0.1	1.3	1.4	1.2	1.3	1.1	1.5	1.2	1.1		
	PrecStd	3.8	0.9	0.7	1.1	0.8	1.1	0.9	1.1	1.1	1.1	1.1	1.1	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.3	34.1	42.0	61.8	80.9	87.4	87.9	82.6	73.0	58.7	39.5	32.3		
		MCWB	31.7	32.7	34.8	49.4	59.3	64.2	67.7	68.5	60.4	50.9	38.4	31.6		
	2%	DB	26.4	28.7	37.5	54.0	73.7	82.8	84.3	78.4	67.3	52.2	35.7	29.9		
		MCWB	25.8	27.8	32.6	44.6	56.3	63.7	66.3	66.0	57.7	47.3	34.8	29.3		
	5%	DB	21.5	24.6	34.2	48.6	67.8	79.2	81.0	73.8	62.5	48.6	33.5	26.6		
		MCWB	20.7	23.6	31.3	41.3	54.4	63.0	65.1	63.2	55.0	45.1	32.9	26.1		
	10%	DB	17.5	20.9	32.0	44.7	61.0	75.1	77.4	70.0	58.7	45.3	31.5	22.7		
		MCWB	16.8	19.8	29.7	38.9	50.7	61.6	64.1	61.6	53.3	42.4	30.7	21.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.6	33.1	35.6	50.3	61.2	69.0	70.9	70.3	62.0	51.8	38.5	31.9		
		MCDB	32.2	34.1	39.4	60.2	75.5	78.5	80.5	79.0	70.0	56.1	39.3	32.3		
	2%	WB	25.6	27.9	33.5	45.9	57.8	66.6	69.2	67.3	58.8	48.7	34.8	28.9		
		MCDB	26.1	28.6	36.0	52.8	70.3	77.0	78.9	74.8	65.3	52.0	35.3	29.3		
	5%	WB	20.7	23.6	31.9	42.1	54.9	64.9	67.5	65.1	55.9	45.4	32.6	25.9		
		MCDB	21.3	24.3	34.1	47.4	65.9	75.4	76.7	71.8	61.1	48.2	33.0	26.5		
	10%	WB	16.7	19.7	29.4	39.2	51.7	62.8	65.7	62.8	53.5	42.5	30.2	21.8		
		MCDB	17.3	20.6	31.9	44.3	61.2	73.3	75.0	68.6	58.0	44.9	31.1	22.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.4	14.6	17.0	16.6	17.1	16.8	16.4	14.4	12.3	9.7	11.9	13.1		
		MCDBR	14.3	12.9	15.0	19.5	25.1	22.9	21.2	20.2	18.0	12.9	8.7	13.9		
		MCWBR	14.4	12.6	11.9	12.3	12.0	9.6	8.4	10.0	10.2	8.9	8.5	13.7		
	5% WB	MCDBR	14.6	13.1	14.1	17.9	23.5	20.1	18.7	18.2	16.5	12.2	9.4	13.8		
		MCWBR	14.6	12.8	12.0	12.4	12.9	9.9	8.8	10.5	10.4	9.0	9.4	13.7		
(40) Clear-Sky Solar Irradiance	taub		0.254	0.298	0.351	0.373	0.386	0.400	0.421	0.414	0.357	0.327	0.264	0.237		
	taud		1.960	2.011	1.958	2.072	2.291	2.345	2.277	2.264	2.387	2.238	2.020	1.899		
	Ebn at Noon		133	203	231	255	264	262	252	242	238	195	128	82		
	Edh at Noon		14	25	38	42	36	36	37	35	26	21	13	9		
(44) All-Sky Solar Radiation	RadAvg		114	391	912	1441	1601	1705	1623	1141	691	297	115	54		
	RadStd		10	40	50	146	100	138	168	120	73	32	26	4		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	-662	N/A	N/A	
(47) Regional (5 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	-662	N/A	N/A	

Nomenclature: See separate page